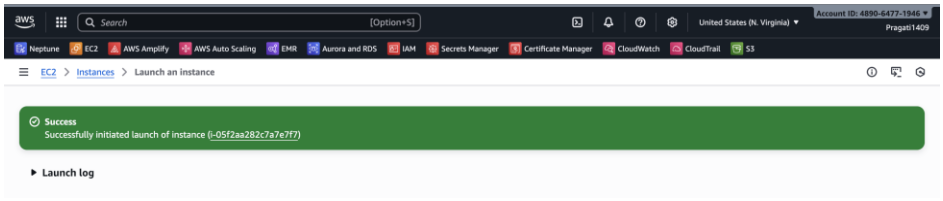
MSML650 – Lab 5

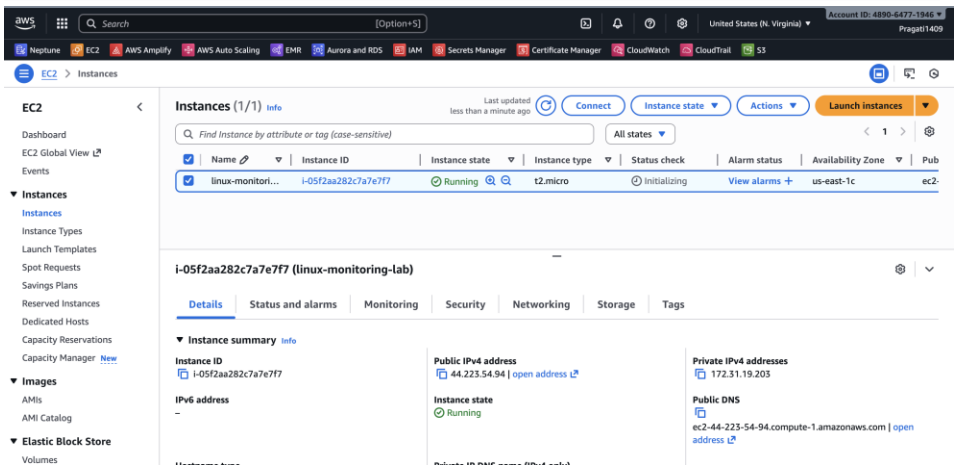
Name: Pragati Rao (UID: 121332776)

Process Monitoring and Management

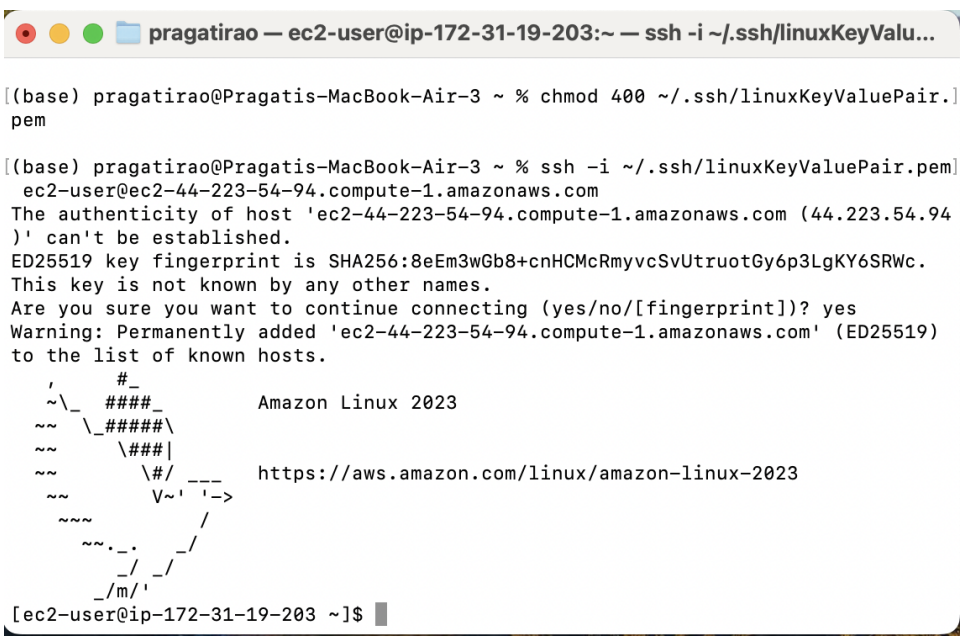
A. Preparation

1. EC2 Instance





2. SSH session



```
pragatirao — ec2-user@ip-172-31-19-203:~ — ssh -i ~/.ssh/linuxKeyValu...

(base) pragatirao@Pragatis-MacBook-Air-3 ~ % chmod 400 ~/.ssh/linuxKeyValuePair.
pem

(base) pragatirao@Pragatis-MacBook-Air-3 ~ % ssh -i ~/.ssh/linuxKeyValuePair.pem
 ec2-user@ec2-44-223-54-94.compute-1.amazonaws.com
The authenticity of host 'ec2-44-223-54-94.compute-1.amazonaws.com (44.223.54.94
)' can't be established.
ED25519 key fingerprint is SHA256:8eEm3wGb8+cnHCMcRmyvcSvUtruotGy6p3LgKY6SRWc.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-44-223-54-94.compute-1.amazonaws.com' (ED25519)
to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-172-31-19-203 ~]$
```

B. Linux Command

3. Linux command 'top'

```
pragatirao — ec2-user@ip-172-31-19-203:~ — ssh -i ~/.ssh/linuxKeyValu...
TOP(1)                          User Commands                          TOP(1)

NAME
       top - display Linux processes

SYNOPSIS
       top -hv|-bcEeHiOSs1 -d secs -n max -u|U user -p pids -o field -w [cols]

       The traditional switches `-' and whitespace are optional.

DESCRIPTION
       The  top program provides a dynamic real-time view of a running system.
       It can display  system  summary  information  as  well  as  a  list  of
       processes  or threads currently being managed by the Linux kernel.  The
       types of system summary information shown and the types, order and size
       of  information  displayed  for processes are all user configurable and
       that configuration can be made persistent across restarts.

       The program  provides  a  limited  interactive  interface  for  process
       manipulation  as  well  as a much more extensive interface for personal
       configuration  --  encompassing every aspect  of  its  operation.   And
       while top is referred to throughout this document, you are free to name
       the program anything you wish.  That new name, possibly an alias,  will
 Manual page top(1) line 1 (press h for help or q to quit)
```

4. Available disk space in your instance

```
pragatirao — ec2-user@ip-172-31-19-203:~ — ssh -i ~/.ssh/linuxKeyValu...
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-44-223-54-94.compute-1.amazonaws.com' (ED25519)
to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-172-31-19-203 ~]$ man top
[ec2-user@ip-172-31-19-203 ~]$ df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M     0  4.0M   0% /dev
tmpfs           475M     0  475M   0% /dev/shm
tmpfs           190M  444K  190M   1% /run
/dev/xvda1      8.0G  1.6G  6.4G  20% /
tmpfs           475M     0  475M   0% /tmp
/dev/xvda128     10M  1.3M  8.7M  13% /boot/efi
tmpfs            95M     0   95M   0% /run/user/1000
[ec2-user@ip-172-31-19-203 ~]$
```

5. Install Python 3 on your instance

```
pragatirao — ec2-user@ip-172-31-19-203:~ — ssh -i ~/.ssh/linuxKeyValu...
  ~~      \###|
  ~~        \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~        V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-172-31-19-203 ~]$ man top
[ec2-user@ip-172-31-19-203 ~]$ df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M     0  4.0M   0% /dev
tmpfs           475M     0  475M   0% /dev/shm
tmpfs           190M  444K  190M   1% /run
/dev/xvda1      8.0G  1.6G  6.4G  20% /
tmpfs           475M     0  475M   0% /tmp
/dev/xvda128     10M  1.3M  8.7M  13% /boot/efi
tmpfs            95M     0   95M   0% /run/user/1000
[ec2-user@ip-172-31-19-203 ~]$ sudo yum install -y python3
Amazon Linux 2023 Kernel Livepatch repository   275 kB/s |  29 kB     00:00
Package python3-3.9.24-1.amzn2023.0.4.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-19-203 ~]$
```

6. **Install the nano text editor on your instance**

```
pragatirao — ec2-user@ip-172-31-19-203:~ — ssh -i ~/.ssh/linuxKeyValu...
       _/m/'
[ec2-user@ip-172-31-19-203 ~]$ man top
[ec2-user@ip-172-31-19-203 ~]$ df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M     0  4.0M   0% /dev
tmpfs           475M     0  475M   0% /dev/shm
tmpfs           190M  444K  190M   1% /run
/dev/xvda1      8.0G  1.6G  6.4G  20% /
tmpfs           475M     0  475M   0% /tmp
/dev/xvda128     10M  1.3M  8.7M  13% /boot/efi
tmpfs            95M     0   95M   0% /run/user/1000
[ec2-user@ip-172-31-19-203 ~]$ sudo yum install -y python3
Amazon Linux 2023 Kernel Livepatch repository   275 kB/s |  29 kB     00:00
Package python3-3.9.24-1.amzn2023.0.4.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-19-203 ~]$ sudo yum install -y nano
Last metadata expiration check: 0:01:05 ago on Mon Nov 17 22:00:04 2025.
Package nano-8.3-1.amzn2023.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-19-203 ~]$
```

7. **Check the available disk space in your instance again. Answer the Questions?**

```
pragatirao — ec2-user@ip-172-31-19-203:~ — ssh -i ~/.ssh/linuxKeyValu...
/dev/xvda128     10M  1.3M  8.7M  13% /boot/efi
tmpfs            95M     0   95M   0% /run/user/1000
[ec2-user@ip-172-31-19-203 ~]$ sudo yum install -y python3
Amazon Linux 2023 Kernel Livepatch repository   275 kB/s |  29 kB     00:00
Package python3-3.9.24-1.amzn2023.0.4.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-19-203 ~]$ sudo yum install -y nano
Last metadata expiration check: 0:01:05 ago on Mon Nov 17 22:00:04 2025.
Package nano-8.3-1.amzn2023.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-19-203 ~]$ df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M     0  4.0M   0% /dev
tmpfs           475M     0  475M   0% /dev/shm
tmpfs           190M  444K  190M   1% /run
/dev/xvda1      8.0G  1.6G  6.4G  20% /
tmpfs           475M     0  475M   0% /tmp
/dev/xvda128     10M  1.3M  8.7M  13% /boot/efi
tmpfs            95M     0   95M   0% /run/user/1000
[ec2-user@ip-172-31-19-203 ~]$
```

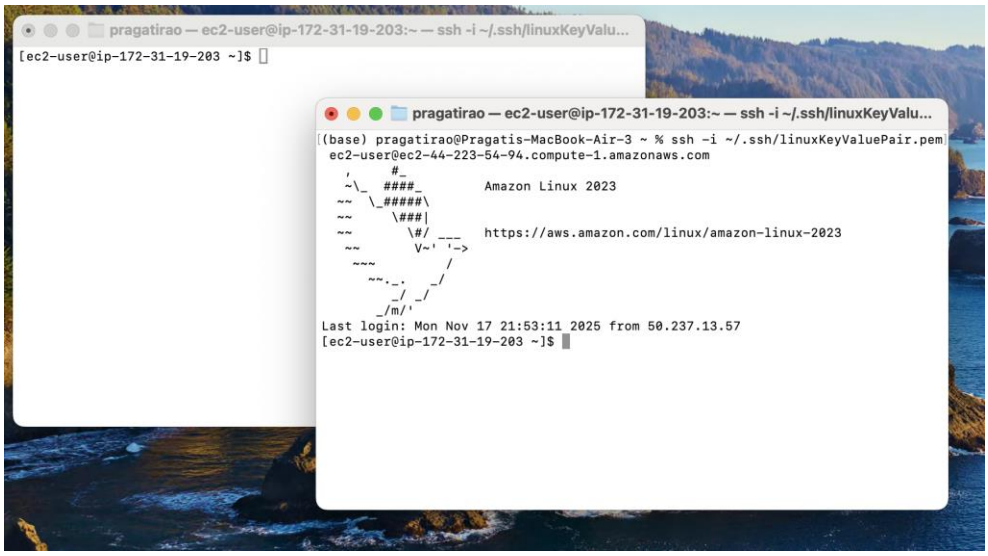
Answer: There is **no difference** in the disk space values. The **Used** value on /dev/xvda1 remained **1.6G**, and the **Available** space remained **6.4G**. Based on the df -h output, they appear to take up **effectively 0 GB** (no measurable change at the 0.1G level). And yes, this is consistent, because the installed sizes were small enough that df -h (which rounds to the nearest 0.1G) did not show any change.

8. **Python program using nano and save as test.py**

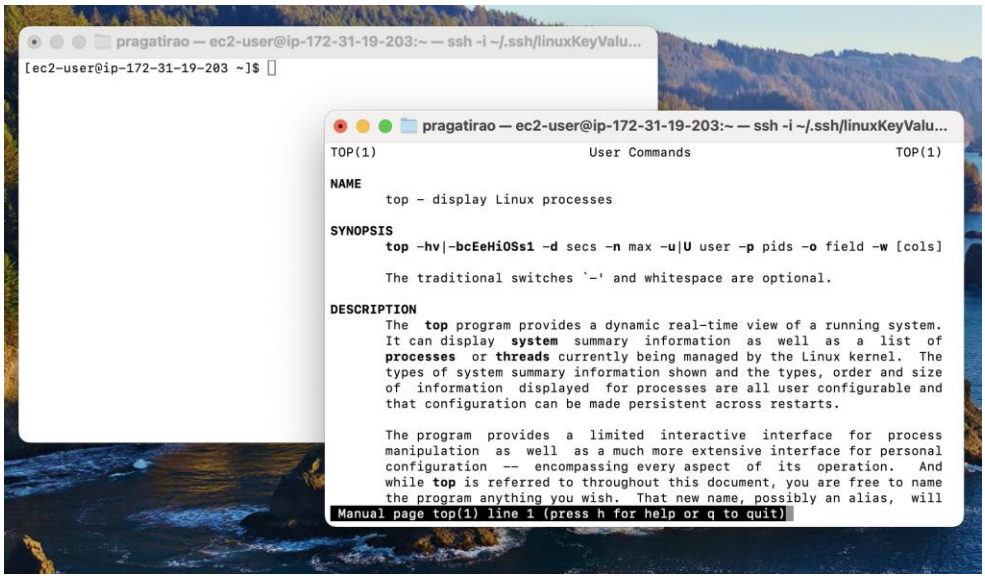
```
pragatirao — ec2-user@ip-172-31-19-203:~ — ssh -i ~/.ssh/linuxKeyValu...
  GNU nano 8.3                    test.py                    Modified
# simple Python program
while True:
    sum = 0
    for n in range(1000000):
        sum += n
    print("The sum is: {0}".format(sum))

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line
```

```
[ec2-user@ip-172-31-19-203 ~]$ nano test.py
[ec2-user@ip-172-31-19-203 ~]$ python3 test.py
The sum is: 499999500000
The sum is: 499999500000
The sum is: 499999500000
The sum is: 499999500000
The sum is: 499999500000
The sum is: 499999500000
The sum is: 499999500000
The sum is: 499999500000
The sum is: 499999500000
The sum is: 499999500000
The sum is: 499999500000
The sum is: 499999500000
The sum is: 499999500000
The sum is: 499999500000
The sum is: 499999500000
```

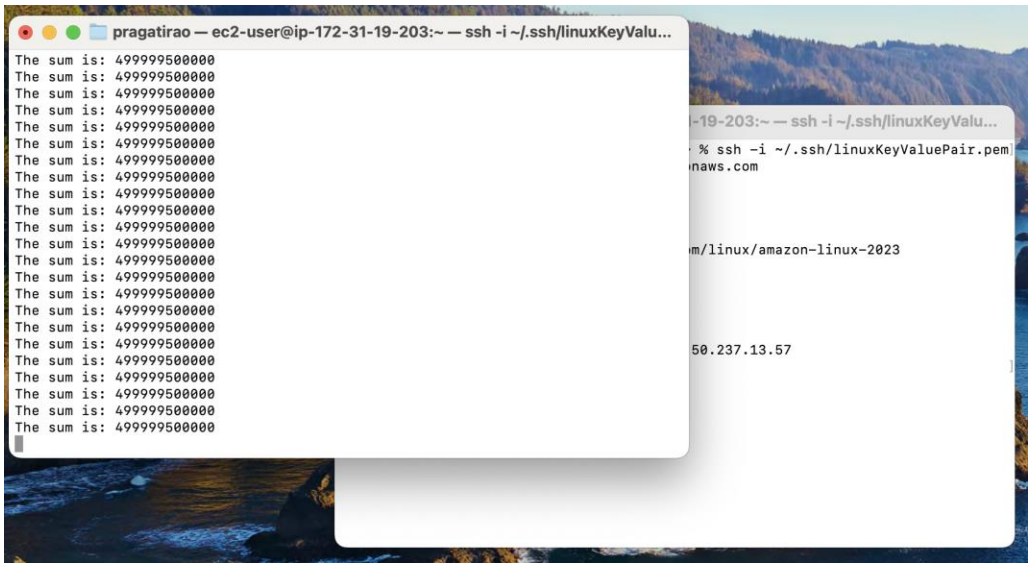
9. **Second SSH session**



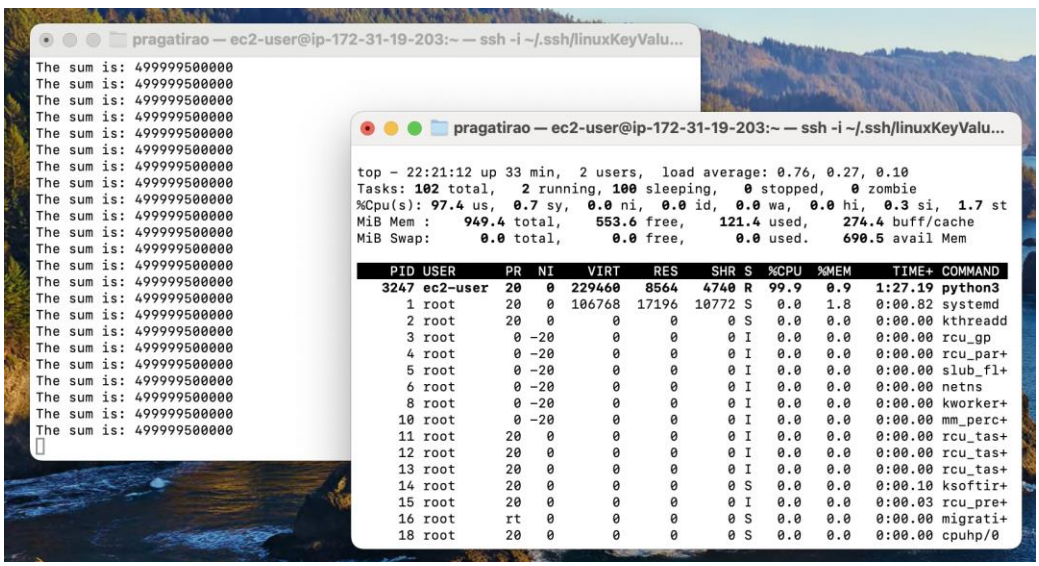
10. **2nd SSH window, use the 'top' command**



11. **1st SSH window, start the python script (type: python3 test.py)**



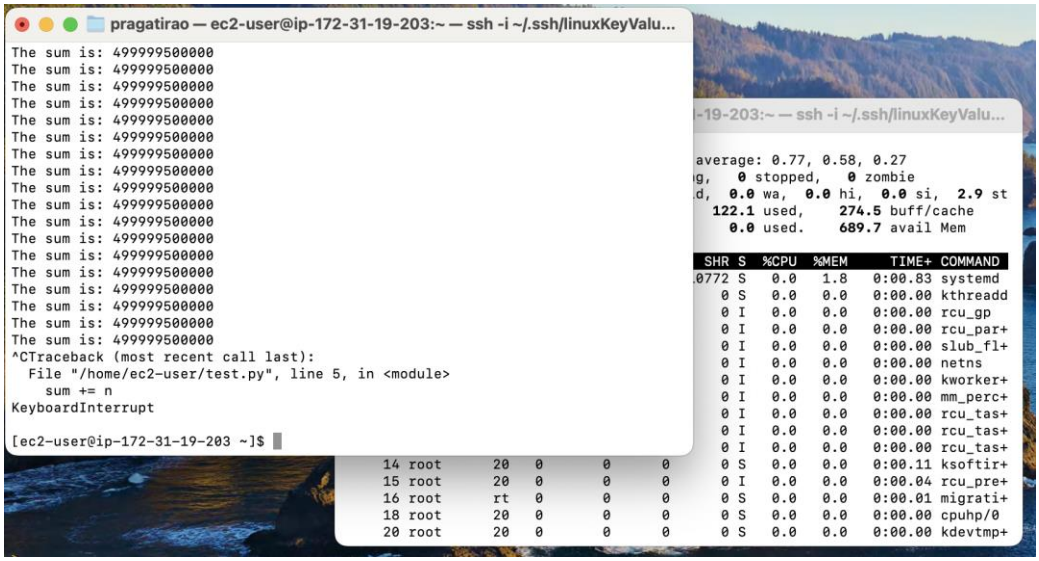
12. CPU utilization in 2nd window

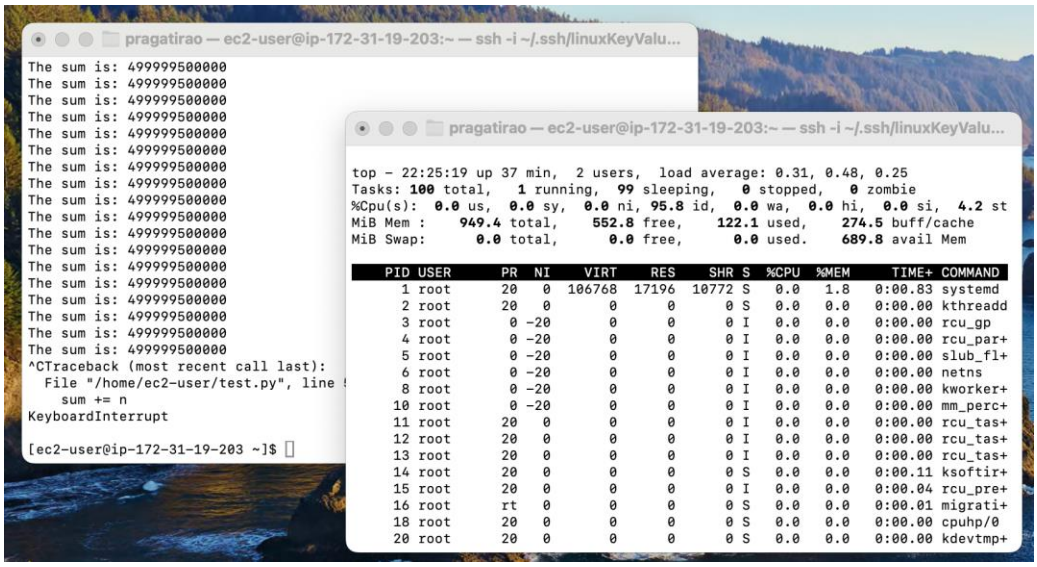


Answer:

- **Max CPU process:** python3
- **CPU utilization:** 99.9%
- **PID:** 3247
- **Memory usage:** ~8.4 MB

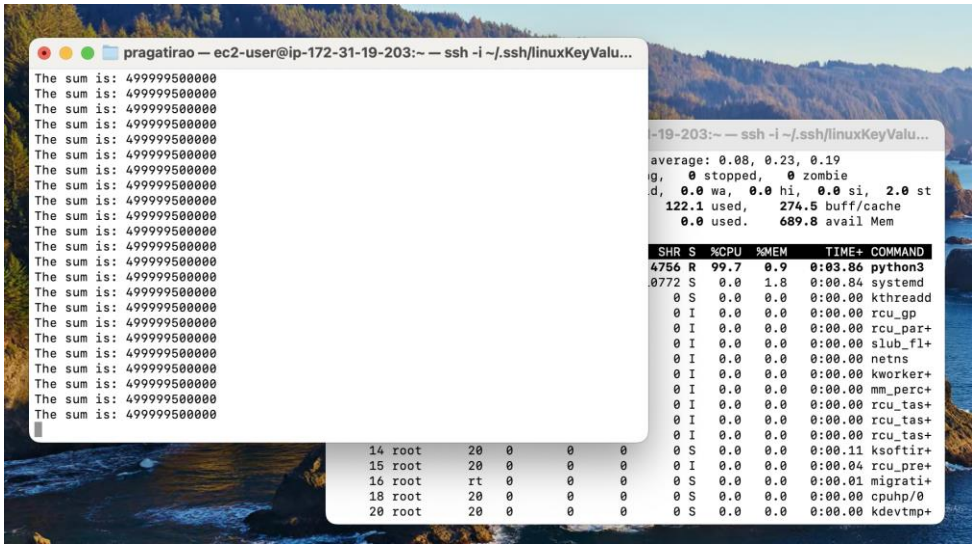
13. 1st SSH window, stop the running Python program



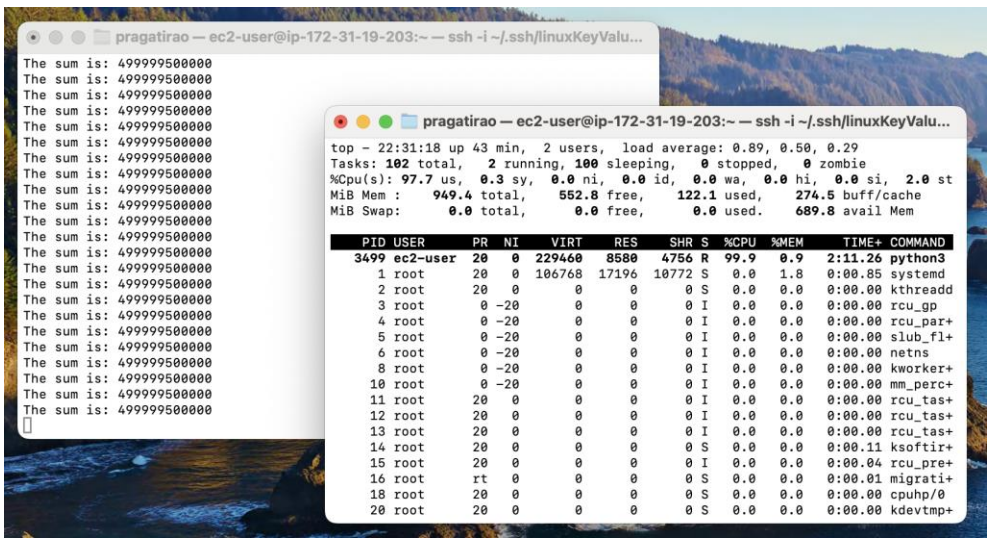


Answer: After stopping the Python program, the overall CPU utilization dropped from nearly 100% to almost 0%. The CPU is now mostly idle (about 95–96% idle).

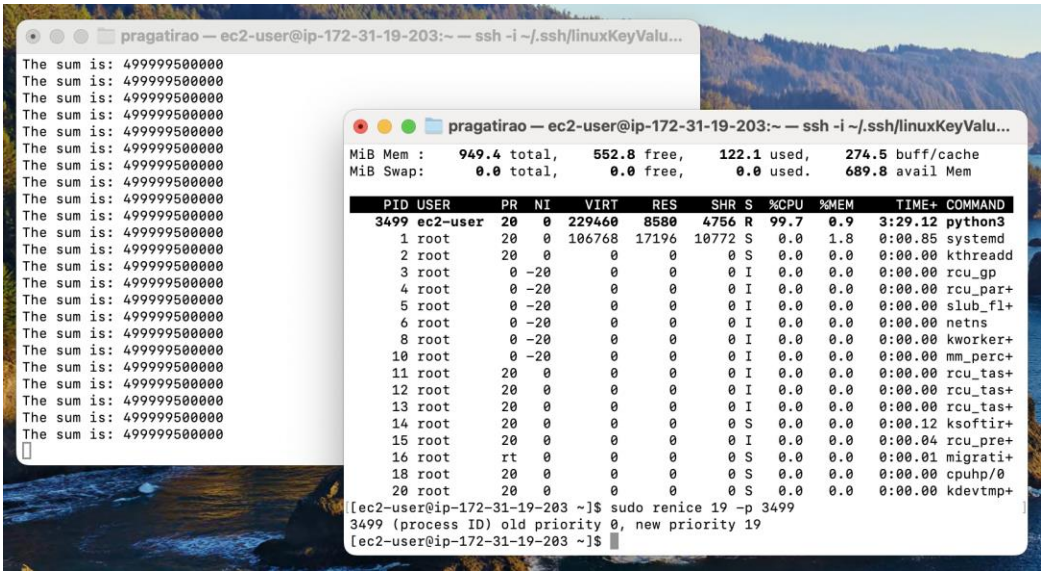
14. 1st SSH window, start the Python script again

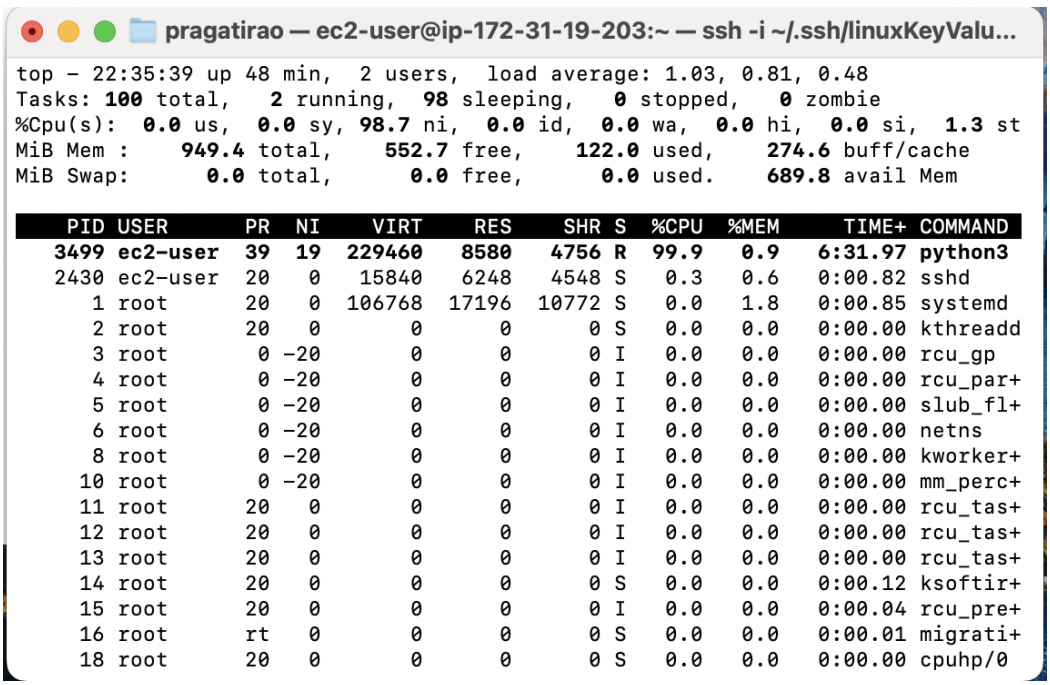


15. 2nd SSH window, note the PID

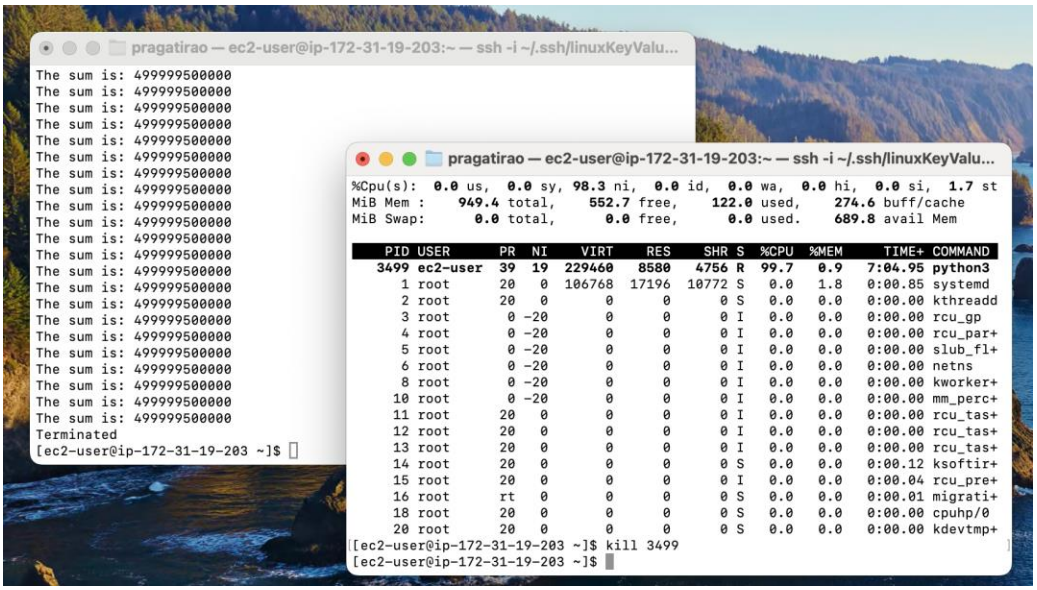


- **PID:** 3499
- **Priority (PR):** 20
- **Nice value (NI):** 0

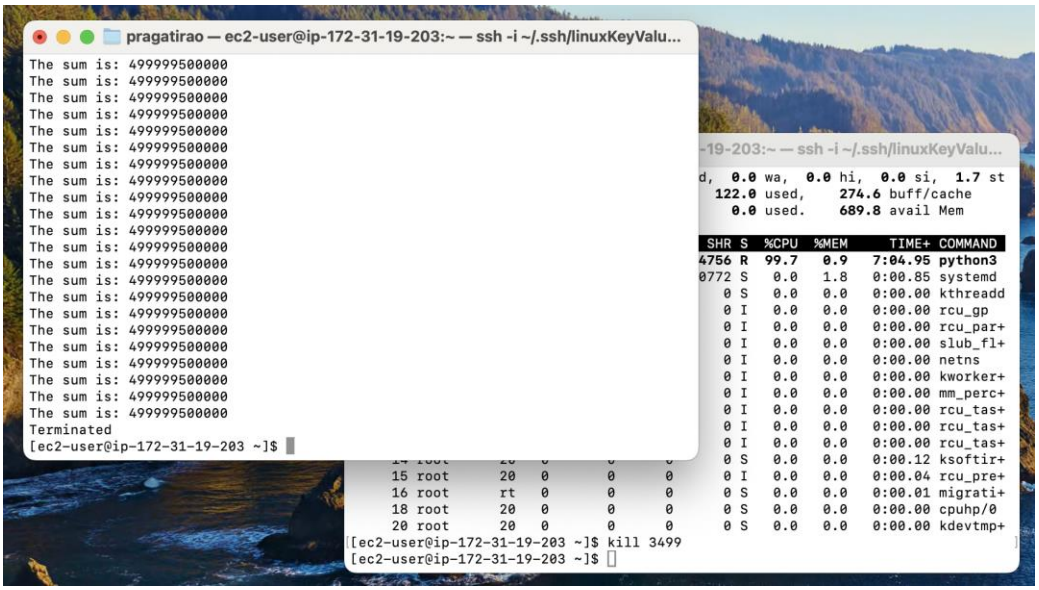




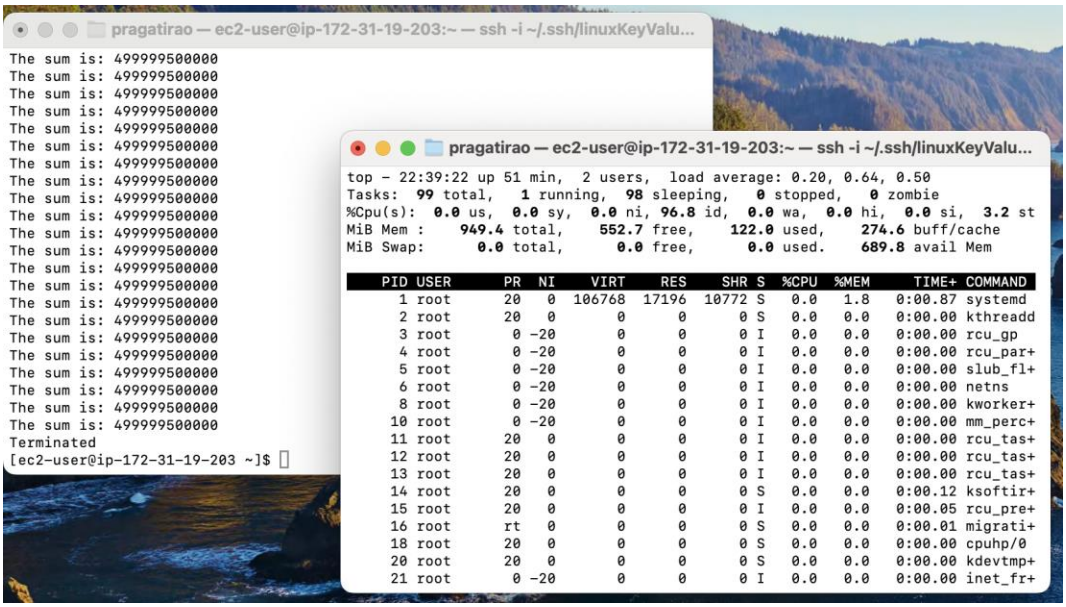
16. 2nd SSH window, terminate the python process



17. 1st SSH window and check if the process was terminated



18. 2nd SSH window, run top again



19. 1st SSH window, delete the Python script

```
pragatirao — ec2-user@ip-172-31-19-203:~ — ssh -i ~/.ssh/linuxKeyValu...
The sum is: 499999500000
The sum is: 499999500000
The sum is: 499999500000
The sum is: 499999500000
The sum is: 499999500000
The sum is: 499999500000
The sum is: 499999500000
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The sum is: 499999500000
The sum is: 499999500000
The sum is: 499999500000
The sum is: 499999500000
The sum is: 499999500000
The sum is: 499999500000
Terminated
[ec2-user@ip-172-31-19-203 ~]$ rm test.py
[ec2-user@ip-172-31-19-203 ~]$ ls
[ec2-user@ip-172-31-19-203 ~]$
```
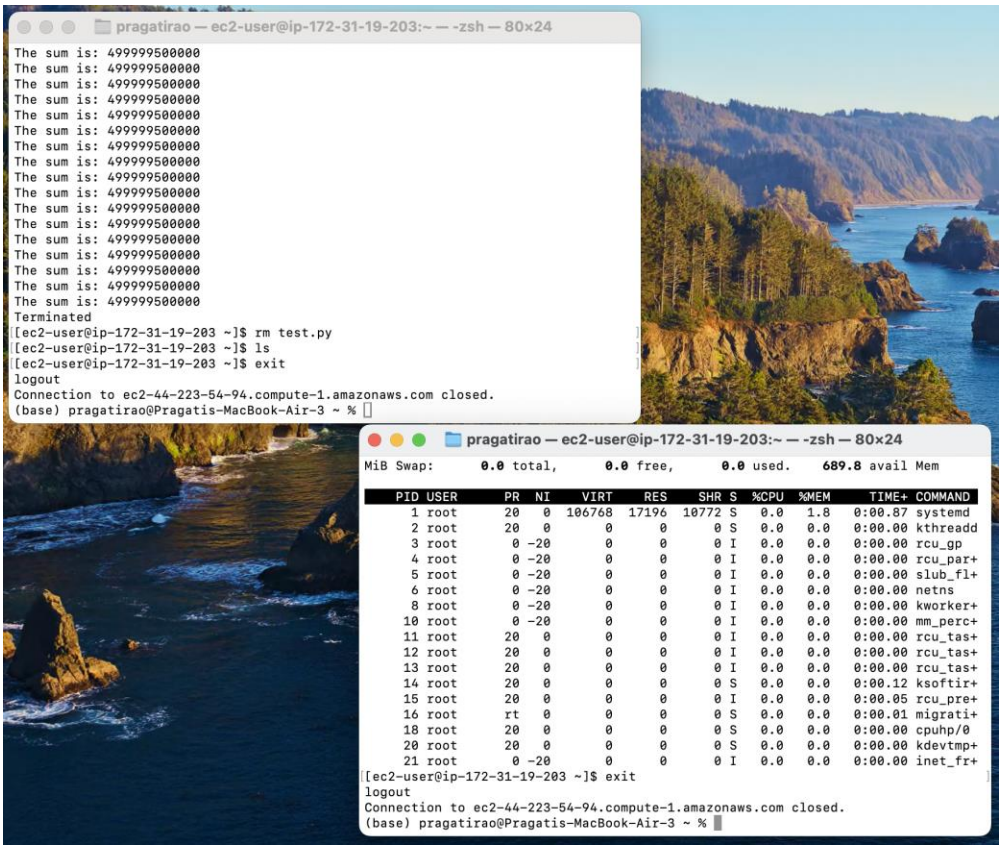
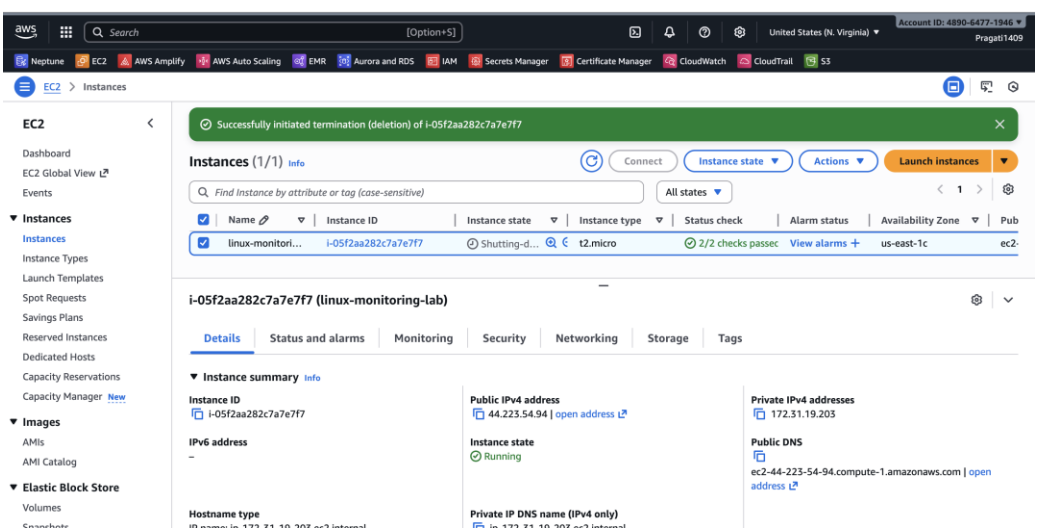

20. Exit from both SSH windows



C. Clean up after yourself

21. Terminate the EC2 instance



22. Log out of AWS